

AI BASED INTEGRATED SOLUTION FOR SMART PEPPER FARMING IN SRI LANKA

25-26J-479

Project Proposal Report

Meegama D M

B.Sc. (Hons) Degree in Information Technology Specialized in
Software Engineering

Department of Computer Science and Software Engineering

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Meegama D M	IT22097724	<u>Madora</u> .

The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor:



Date: 23/08/2025.

Signature of the:
co-supervisor



Date: 23/08/2025

ABSTRACT

Pepper is an important export crop for Sri Lanka, contributing significantly to the agricultural economy. Sri Lanka produces and exports black pepper but Sri Lankan black pepper suppliers satisfy only 2% of the global demand for pepper. However, farmers face challenges in accurately predicting yield during early growth stages, often relying on manual observation and experience. Such methods are subjective, time-consuming, and can lead to errors, resulting in inefficiencies in harvest planning, resource allocation, and market readiness. The lack of modern, data-driven forecasting limits both productivity, correct decision making regarding future and present pepper cultivations and profitability.

This research proposes a machine learning based yield prediction system for pepper plants, leveraging convolutional neural networks (CNNs) trained on images of early stage pepper cones. Environmental, climatic, and geographical location data will be integrated to enhance model accuracy. Once predictions are generated, Explainable Artificial Intelligence (XAI) will be used to get insights per-plant yield directly in the field via a mobile device, enabling farmers for effective decision making. Additionally, plantation-wide heatmaps will be produced from drone and mobile imagery to highlight areas of varying yield intensity.

The proposed system is designed to empower farmers with actionable insights, improve harvest forecasting accuracy, and enable more efficient labor and input management. By integrating Machine Learning(ML), Internet of Things(IoT) and Explainable AI (XAI) addresses existing gaps in early yield estimation, offering a scalable, user-friendly, and innovative approach to advancing Sri Lanka's pepper industry .

Key Words- Machine Learning, Internet of Things , Yield Prediction, Explainable AI

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LIST OF ABBREVIATIONS

Abbreviation	Description
AR	Augmented Reality
ML	Machine Learning
CNN	Convolutional Neural Network
AI	Artificial Intelligence

Sub-Topic: IoT-Driven Yield Prediction System for Black Pepper Cultivation in Sri Lanka
Using Machine Learning and Explainable AI

1. INTRODUCTION

Pepper (*Piper nigrum L.*) is one of Sri Lanka's most valuable export crops, which is also considered as King of Spices contributing significantly to the country's agricultural economy. However, the industry faces critical challenges such as unpredictable yields, inconsistent quality grading, lack of transparency in the supply chain, and limited access to advanced farming technologies for smallholder farmers. At present, most yield estimations rely on farmers' experience and visual inspection, which can be subjective, inaccurate, and delayed. This leads to inefficient resource allocation, unexpected harvesting schedules, and missed market opportunities.

Advancements in Artificial Intelligence , Machine Learning present an opportunity to modernize pepper cultivation. Machine learning models, particularly those based on computer vision, can analyze images of pepper plants at early fruiting stages, environmental factors, climatic factors and geographical location data to predict final yield more accurately.

A novel aspect of this research is the integration of XAI to gain insights about factors affecting yield of pepper. Traditional statistical yield estimation techniques and farmer intuition often fail to capture these complex, non-linear relationships between soil conditions, water levels, nutrient levels and sunlight conditions. Consequently, farmers struggle to plan fertilizer use, irrigation, and harvest timing efficiently. Farmers can point a mobile device at their plants and placing the IoT device near the pepper plant can help farmer to get the expected yield and insights based on XAI. This approach provides an intuitive understanding of crop development, empowering farmers to plan harvesting and market strategies effectively.

The proposed system will leverage,

- Machine Learning (CNN models) for image-based yield prediction from young pepper cone photographs.
- (XAI) techniques to accurately and transparently predict black pepper yield in Sri Lanka.

By combining predictive analytics with immersive visualization, this research aims to deliver a practical, farmer-friendly solution that bridges the gap between advanced agricultural technology and real-world farming needs in Sri Lanka.

1.1 Background and Literature Review

Pepper (*Piper nigrum L.*) is one of Sri Lanka's most valuable export crops, playing a crucial role in the agricultural economy and supporting rural livelihoods [1]. The island's tropical climate and fertile soils create favorable conditions for pepper cultivation. However, despite these natural advantages, productivity levels remain inconsistent. According to data from the *Industry Data Book* published by the Ministry of Industries [4], Sri Lanka's pepper exports have shown a noticeable decline between 2020 and 2022. This downward trend is illustrated in the diagram below.

Table 2: Spices Products Export - (2020-2022)

Description	2020		2021		2022	
	Qty (Kg)	Value (Rs.)	Qty (Kg)	Value (Rs.)	Qty (Kg)	Value (Rs.)
Pepper of the genus piper, dried or crushed or ground fruits of the genus capsicum or of the genes Pimenta	9,503,371	9,430,817,245	18,493,862	23,054,745,272	11,712,371	23,902,479,575
Vanilla: Neither crushed nor ground: Vanilla pods	624	25,726,958	1,432	30,459,577	853	49,016,302
Cinnamon and cinnamon-tree flowers	18,721,832	38,218,806,221	18,813,519	45,879,168,502	18,298,194	71,716,168,633
Cloves (whole fruit, cloves and stems)	2,579,388	2,235,343,109	6,478,676	6,888,136,089	2,325,133	6,563,803,597
Nutmeg, mace and cardamon	2,047,549	3,046,324,689	1,992,982	3,578,891,529	2,286,554	6,157,633,413
Seeds of anise, badian, fennel, coriander, cumin or caraway; juniper berries	84,262	29,313,139	37,921	26,700,936	23,286	27,834,489
Ginger, saffron, turmeric (curcuma), thyme, bay leaves, curry and other spices	1,278,102	1,047,490,164	1,403,137	1,386,706,968	1,116,819	1,710,619,756
Grand Total	34,215,128	54,033,821,525	47,221,530	80,844,808,873	35,763,210	110,127,555,765

Source: Sri Lanka Customs

Figure 1: Pepper Export from 2020-2022

Smallholder farmers, who represent the majority of producers, face unpredictable yields due to pest and disease outbreaks, fluctuating weather conditions, and the absence of scientific yield estimation tools [2].

At present, yield prediction in pepper farming largely relies on manual observation and farmer experience. This traditional approach is subjective, time-consuming, and often inaccurate particularly during the early stages of fruit development when accurate forecasting would be most beneficial. Inaccurate yield estimation leads to poor harvest planning, insufficient allocation of processing capacity, and unreliable market supply commitments, all of which negatively affect farmer income and market stability [3].

Recent advancements in Machine Learning offer promising opportunities to address these challenges. By providing accurate yield estimates through models it can acts as valuable tools for farmers to better plan and manage black pepper cultivation with different environmental conditions. Smart agriculture has emerged as a crucial approach to improving agricultural productivity, resource optimization, and risk reduction in farming activities. Researchers highlight that traditional farming practices often rely on farmers' experience and manual observations, which limits accuracy and consistency in decision-making. Recent advancements in the Internet of Things (IoT) and Machine Learning (ML) allow the development of intelligent systems capable of collecting real-time environmental data and making accurate predictions. These innovations have proven effective in yield forecasting, disease detection, and soil monitoring across a variety of crops.

IoT systems have become widely used to monitor key agricultural parameters such as soil moisture, temperature, humidity, solar radiation, and rainfall. Studies demonstrate that IoT-based sensors combined with cloud platforms significantly improve the timeliness and accuracy of data collection for Sri Lankan agriculture. Similar systems developed for paddy and tea plantations show improvements in water management and soil health monitoring. However, most of the existing IoT studies either rely on static datasets or do not integrate predictive analytics to generate actionable insights for farmers. Moreover, IoT-based research on spice crops, especially black pepper, remains limited.

This study [5] presents a hybrid ML model with IoT for yield prediction. This involves three phases preprocessing, feature selection and classification. This paper[5] proposed an IoT-enabled

crop yield prediction model that integrates soil sensors with a two-tier machine learning architecture for predictive analytics. Their framework uses moisture, temperature–humidity, and pH sensors connected through an Arduino–Raspberry Pi IoT pipeline, after which soil-quality features are processed using Correlation-Based Feature Selection (CBFS) and Variance Inflation Factor (VIF) methods. The first tier employs an Adaptive k-Nearest Centroid Neighbour Classifier (aKNCN) for soil classification, while the second tier utilizes an Extreme Learning Machine optimized with a Modified Butterfly Optimization Algorithm (ELM-mBOA) for yield regression. The model achieved strong performance ($RMSE = 0.301$; $R^2 = 0.817$) on an Indian soil dataset, demonstrating the usefulness of IoT-driven data collection for yield prediction. However, the study focuses solely on soil-chemical parameters and does not incorporate crop images, or explainable AI components. Furthermore, the model is not crop-specific and therefore does not address the physiological and environmental characteristics required for specialty crops such as black pepper. These gaps highlight the need for an IoT and XAI-driven, crop-specific modeling framework tailored to Sri Lankan agricultural conditions.

Siam *et al.* [6] proposes a next-generation IoT-based device designed to predict crop yield by integrating real-time sensor measurements with multiple machine learning models. The device collects field data such as soil moisture, temperature, humidity, and irrigation status through an IoT system comprising a soil moisture sensor, GSM module, solar-powered controller, and automated drip-irrigation mechanism. The collected data is fed into a hybrid ML framework combining Artificial Neural Networks (ANN), Fuzzy Logic, and Support Vector Machines (SVM). ANN is used for prediction tasks such as soil temperature and yield-related variables, Fuzzy Logic handles uncertainty in environmental conditions, while SVM is applied for classification tasks including soil type identification and plant infection detection.

The proposed system demonstrates strong predictive performance, with ANN achieving the highest accuracy ($R^2 = 0.94$) and lowest mean squared error compared to Fuzzy Logic and SVM. The framework also incorporates rainfall prediction using ANN and uses image-based SVM for crop health classification, achieving up to 96% accuracy for detecting plant infections. Field deployment results show effective real-time monitoring and automated irrigation control, enhancing water efficiency and supporting better crop management decisions. Overall, the study

highlights the potential of combining IoT and ML for precision agriculture, though it focuses mainly on general environmental parameters rather than crop-specific physiological features.

Talaat *et al.* [7] presents the CYPA framework as a robust IoT- and machine-learning-based approach for predicting national-scale crop yields, the system remains largely focused on macro-level climate indicators—such as average temperature, rainfall, and pesticide usage—and does not integrate fine-grained, crop-specific physiological data. The models used (Decision Tree Regressor, Random Forest Regressor, Extra Tree Regressor) achieve high accuracy at a global agricultural scale, yet they rely exclusively on tabular climatic and agricultural datasets, limiting the framework’s ability to capture plant-level variations, growth stages, or disease symptoms that strongly influence yield in specialty, perennial crops like black pepper. Furthermore, CYPA’s outputs, although accurate, remain non-interpretable, as no Explainable AI (XAI) techniques are incorporated to clarify the influence of different environmental or agronomic variables on model predictions. In contrast, the proposed system for Sri Lankan black pepper cultivation integrates IoT sensor data, plant images, agronomic parameters, and deep learning architectures to generate more localized and crop-specific predictions. Moreover, the inclusion of XAI mechanisms such as SHAP, LIME, or Grad-CAM offers interpretability that is critical for farmer trust, advisory services, and decision-making. Additionally, CYPA’s models are developed using global datasets that may not reflect Sri Lanka’s unique microclimates, soil structures, and pepper-specific cultivation practices, whereas the proposed black-pepper model is designed to adapt to real field data, local agronomic conditions, and the phenology of the pepper plant. Therefore, while CYPA contributes an important foundation for climate-aware yield forecasting, it does not address the requirements of crop-specific, explainable, and multimodal prediction systems needed for high-value crops such as black pepper.

Many researchers have explored ML algorithms to predict crop yields. Commonly used models include Random Forest, Support Vector Regression (SVR), XGBoost, Linear Regression, and Artificial Neural Networks.

Subasinghe and Kulathilake proposed a climate-driven machine learning approach to predict black pepper yield and quality in Sri Lanka using a Long Short-Term Memory (LSTM) model[8]. The study utilized 30 years of historical climatic data from the Matale district, including rainfall, rainy days, drought duration, temperature, and humidity, alongside agronomic yield and quality indicators such as bulk density, piperine content, and oleoresin content. Correlation analysis revealed that rainfall-related parameters—annual rainfall, number of rainy days, and rainfall following drought periods—exhibited the strongest influence on both yield and quality, while temperature and relative humidity showed comparatively weaker effects. Based on these findings, separate LSTM models were developed for yield and quality prediction. The models demonstrated reasonable predictive performance, achieving mean absolute errors of approximately 18–20% and improved accuracy when simulated data were incorporated to mitigate data scarcity.

This study is significant as it represents one of the first machine-learning-based attempts to predict black pepper yield and quality in Sri Lanka, highlighting the suitability of LSTM networks for capturing long-term temporal dependencies between climatic variables and crop performance. However, the authors note limitations related to the exclusion of soil characteristics and farm management practices, suggesting opportunities for future model enhancement and broader regional applicability.

Karthik *et al.* [9] developed a crop weather analysis model (CWAM) to estimate black pepper yield by examining the relationship between climatic variables and crop productivity. Using seven years of weather data from West Bengal, India, the study applied second-degree multiple regression models incorporating variables such as rainfall, maximum and minimum temperature, relative humidity, wind speed, evaporation, and sunshine hours. The results revealed that relative humidity (morning and afternoon) and wind speed had the strongest positive influence on yield, while excessive temperature and evaporation negatively affected productivity. The model achieved a high prediction accuracy, with rainfall-based regression explaining up to 90% of yield variation. This study highlights the importance of weather-driven statistical models for

estimating black pepper yield, while also indicating limitations in handling long-term temporal dependencies and complex nonlinear interactions compared to modern machine learning approaches.

Krishnamurthy *et al.* [10] investigated long-term trends in climate variables and their relationship with black pepper (*Piper nigrum*) productivity across major pepper-growing regions in India, including Kerala, Tamil Nadu, and Karnataka. Using two decades of climatic and yield data, the study revealed a general declining trend in rainfall alongside an increase in both maximum and minimum temperatures, which coincided with a decline in black pepper productivity in most regions.

Correlation analysis showed that rainfall during December–January negatively affected yield, while April–May rainfall had a positive influence, highlighting the importance of moisture availability during the flowering and early growth stages. Temperature effects varied by elevation: in higher elevations, minimum temperature showed a positive correlation with productivity, whereas in lowland (plain) regions, both maximum and minimum temperatures negatively impacted yield. Regression analysis further indicated that temperature variables exerted a stronger influence on productivity than rainfall or rainy days.

The study also developed normalized yield deviation (NYD) models to predict future productivity under projected climate scenarios. Results suggested that while some high-elevation areas (e.g., Wayanad) may experience slight yield increases due to rising minimum temperatures, overall climate change—particularly increased temperatures and altered rainfall distribution—poses a significant risk to sustainable black pepper production. The authors emphasize that future prediction models should incorporate additional factors such as soil moisture, humidity, crop management practices, and economic variables to improve prediction accuracy.

Explainable AI is increasingly used to interpret model decisions and improve user trust, especially in high-stake domains like agriculture. Tools such as SHAP (Shapley Additive Explanations) and LIME (Local Interpretable Model-Agnostic Explanations) help identify which environmental features—such as soil moisture or temperature—strongly influence yield predictions. Most existing research focuses purely on improving accuracy without providing transparency for non-technical users.

De Alwis *et al.* [11] propose an explainable artificial intelligence (XAI) framework for crop yield prediction in smart farming environments to address the lack of transparency in deep learning-based agricultural models. The study introduces the Duo Attention eXplainable Mechanism (DAXM), which integrates dual attention mechanisms to capture both feature importance and temporal dependencies in multivariate time-series yield data. The model was evaluated using real-world smart farm datasets for tomato and strawberry yield prediction collected from controlled agricultural environments in South Korea.

Experimental results demonstrate that DAXM not only achieves superior prediction accuracy compared to traditional models such as LSTM, SVR, and XGBoost, but also provides more consistent and interpretable explanations than popular post-hoc explainability techniques including LIME and SHAP. By analyzing a three-week temporal window prior to harvest, the proposed approach identifies critical environmental and plant-related factors—such as temperature, humidity, water availability, flowering rate, and fruit count—that significantly influence yield outcomes. Statistical tests and expert evaluations further confirm that DAXM’s explanations align more closely with agronomic knowledge, thereby enhancing trust and usability among farmers and domain experts.

This work highlights the importance of incorporating explainability into AI-driven yield prediction systems and demonstrates that attention-based deep learning models can effectively balance predictive performance with interpretability, making them more suitable for real-world smart farming applications.

Venugopal *et al.* [12] conducted a comparative evaluation of explainable artificial intelligence (XAI) methods for improving transparency in deep learning-based crop yield estimation using satellite imagery. The study focused on soybean yield prediction using a Convolutional Neural Network (CNN) trained on Sentinel-2 satellite data covering major soybean-producing regions in the United States. To address the black-box nature of deep learning models, multiple XAI techniques—namely Layer-wise Relevance Propagation (LRP), Deep Taylor Decomposition, Grad-CAM, and SmoothGrad—were applied to generate saliency maps that explain model predictions.

A perturbation analysis was employed to quantitatively assess the reliability of each XAI method by measuring changes in prediction accuracy when important image regions were systematically

altered. The results demonstrated that LRP consistently produced more accurate and meaningful explanations, as evidenced by steeper declines in model accuracy when highly relevant regions were perturbed and lower Area Under the Curve (AUC) values compared to other methods. Furthermore, LRP was shown to better focus on soybean cultivation areas rather than non-crop regions, indicating that it captures agriculturally relevant spatial information more effectively.

Overall, the study highlights the importance of selecting appropriate XAI techniques for crop yield estimation tasks and concludes that LRP is more suitable for explaining CNN-based yield predictions using remote sensing data. These findings support the integration of explainable deep learning models in agricultural decision-support systems by improving model transparency, trust, and interpretability.

Pant *et al.* conducted a comparative study on crop yield prediction using both traditional machine learning models and Explainable Artificial Intelligence (XAI) techniques to address the transparency issues of black-box models in agriculture [13]. The study evaluated four regression models—Linear Regression, Decision Tree, Random Forest, and XGBoost—using a real-world agricultural dataset comprising weather, soil, and crop parameters. Model performance was assessed using RMSE, MAE, and R^2 metrics, where XGBoost demonstrated superior predictive accuracy with the lowest RMSE (1.46) and highest R^2 (0.95).

To enhance model interpretability, the authors integrated SHAP and LIME as XAI techniques. SHAP was used to provide global feature importance, revealing that temperature, rainfall, and soil nutrients were the most influential factors affecting crop yield, while LIME offered localized explanations for individual predictions, particularly useful in analyzing outliers. The results indicate that combining high-performing models such as XGBoost with XAI methods significantly improves both prediction accuracy and transparency. This study highlights the importance of explainable and interpretable machine learning approaches for building trustworthy crop yield prediction systems that can support informed decision-making in agriculture [13].

Kumar and Kumar propose an Explainable AI (XAI)-based crop recommendation framework that integrates soil nutrient profiles and historical crop yield data to enhance transparency and trust in smart agriculture systems[14]. The framework employs a hyperparameter optimization-based grid search algorithm combined with multiple machine learning classifiers, including

Logistic Regression, Decision Tree, Random Forest, KNN, SVM, and Gradient Boosting, to identify the most suitable crop for given soil and climatic conditions. To address the black-box nature of conventional ML models, the study incorporates XAI techniques such as LIME and SHAP, enabling farmers to understand the influence of key factors such as nitrogen (N), phosphorus (P), potassium (K), pH, rainfall, humidity, and temperature on crop recommendations. Experimental evaluation using an Indian agricultural dataset demonstrates that the proposed model significantly outperforms baseline approaches, achieving an accuracy of 99.73%, along with improved precision, recall, and error metrics. The study highlights that integrating XAI not only improves predictive performance but also enhances interpretability and user trust, making the framework suitable for real-world smart farming and decision-support applications.

Celik *et al.* [15] propose an explainable artificial intelligence (XAI) framework for cotton yield prediction by integrating multisource data, including satellite remote sensing products, climatic variables, and soil properties. The study employs an Explainable Boosting Machine (EBM), a glass-box model that provides inherent interpretability while maintaining competitive predictive performance. Using data from major cotton-producing regions in the continental United States between 2017 and 2021, the authors demonstrate that EBM achieves comparable accuracy to black-box models such as XGBoost and LightGBM, while outperforming traditional interpretable models.

The explainability analysis reveals that dynamic features, particularly precipitation, enhanced vegetation index (EVI), and leaf area index (LAI), contribute most significantly to yield prediction, accounting for approximately 78% of overall feature importance. Additionally, feature interactions, especially between soil texture and climatic variables, play a substantial role in improving prediction reliability. The study further shows that accurate cotton yield estimation is possible several months before harvest, highlighting the model's practical applicability for early decision-making in agriculture. Overall, this work underscores the importance of combining multisource data with inherently interpretable models to enhance trust, transparency, and effectiveness in crop yield prediction systems.

The reviewed literature demonstrates that IoT-based data collection and machine learning models significantly improve crop yield prediction accuracy across various crops. However,

most existing systems rely on either sensor-based or climatic data alone, with limited integration of plant-level visual information. While recent studies incorporate deep learning and XAI techniques to enhance interpretability, these approaches are rarely applied together in a unified framework. Furthermore, research on black pepper yield prediction is scarce and primarily focused on climatic or statistical models, lacking real-time IoT integration, image-based analysis, and explainability. These limitations highlight the need for a crop-specific, multimodal, and explainable yield prediction system tailored to Sri Lankan black pepper cultivation.

1.2 Research Gap

A comprehensive review of existing literature reveals significant advancements in crop yield prediction using IoT, machine learning, and explainable artificial intelligence. However, several critical gaps remain, particularly in the context of black pepper cultivation in Sri Lanka.

First, although IoT-based systems have been widely applied in agriculture for crops such as paddy, maize, wheat, tea, and cotton, very limited research has focused on spice crops, especially black pepper. Existing IoT-driven yield prediction frameworks primarily rely on soil and climatic sensor data and do not incorporate plant-level visual features, which are crucial for accurately capturing physiological growth patterns in perennial crops like black pepper.

Second, most machine learning-based yield prediction studies depend on tabular datasets (e.g., rainfall, temperature, soil nutrients) and neglect image-based modeling. While CNN-based approaches have shown strong performance in yield estimation for crops such as soybean and cotton using satellite or aerial imagery, image-based yield prediction from ground-level plant images—particularly early-stage pepper cone images—has not been adequately explored. This limits early forecasting capabilities that are essential for harvest planning and market decision-making.

Third, existing black pepper-related studies predominantly focus on climate-driven statistical or time-series models and are largely confined to macroscale yield analysis. These approaches fail to integrate real-time IoT data, localized farm conditions, and plant phenology, resulting in limited adaptability to smallholder farming environments and regional microclimates in Sri Lanka.

Furthermore, despite the growing adoption of deep learning models for yield prediction, model interpretability remains a major challenge. Most high-performing models operate as black boxes, offering little insight into how environmental, soil, or visual factors influence yield outcomes. Although Explainable AI techniques such as SHAP, LIME, Grad-CAM, and LRP have been successfully applied to crops like wheat, maize, soybean, and cotton, their application in black pepper yield prediction is virtually nonexistent. This lack of explainability restricts farmer trust, practical adoption, and decision support.

Finally, current systems often function as isolated analytical tools rather than integrated, farmer-centric solutions. There is a notable absence of end-to-end systems that combine IoT-based real-time sensing, image-based machine learning, and XAI-driven insights into a single, deployable framework tailored to Sri Lankan black pepper farmers.

Study	Crop Focus	IoT Integration	ML Technique	Image Based Prediction	XAI Used	Crop Specific (Black Pepper)	Sri Lankan Context	Proposed
Gupta & Nahar [5]	General crops	✓	aKNCN, ELM	✗	✗	✗	✗	Relies only on soil parameters; no images or explainability
Siam et al. [6]	General crops	✓	ANN, SVM, Fuzzy	✓ (disease only)	✗	✗	✗	Not crop-specific; lacks interpretability
Talaat [7] (CYPA)	Multi-crop (macro-level)	✓	RF, DT, Extra Trees	✗	✗	✗	✗	Uses aggregated climate data, black-box predictions
Subasinghe & Kulathilake [8]	Black pepper	✗	LSTM	✗	✗	✓	✓	Uses only climatic data; no IoT or XAI

Karthik et al. [9]	Black pepper	✗	Regression	✗	✗	✓	✗	Cannot model nonlinear relationships
De Alwis et al. [11]	Tomato, Strawberry	✓	Attention DL	✗	✓ (DAXM)	✗	✗	Controlled environment; not field-level
Venugopal et al. [12]	Soybean	✗	CNN	✓ (satellite)	✓ (LRP)	✗	✗	Not IoT-based; not applicable to pepper
Proposed System	Black pepper	✓	CNN + ML	✓ (plant images)	✓ (SHAP / Grad-CAM)	✓	✓	Addresses all identified gaps

Table 1: Research Gap

As shown in Table 1, most existing studies focus on either IoT-based data collection or machine learning-based yield prediction, but rarely integrate both with explainable AI. While a few studies address black pepper cultivation, they rely primarily on climatic data and lack real-time IoT sensing, image-based prediction, and interpretability. Moreover, none of the reviewed systems are tailored to Sri Lankan black pepper cultivation while providing transparent, farmer-friendly explanations. These gaps highlight the need for an integrated IoT-driven, machine learning-based, and explainable yield prediction system specifically designed for black pepper farming in Sri Lanka.

1.3 Research Problem

Despite black pepper being one of Sri Lanka's most valuable export crops, local farmers lack access to an integrated, early-stage yield prediction system that is accurate, explainable, and tailored to Sri Lankan cultivation conditions. At present, yield estimation in pepper farming predominantly depends on farmers' experience and visual inspection of vines and pepper cones. These traditional practices are subjective, inconsistent, and often unreliable, particularly during the early fruiting stages when timely predictions are most valuable.

Although recent studies have demonstrated the potential of machine learning, IoT-based monitoring, and explainable AI in agricultural yield prediction, most existing systems are designed for large-scale annual crops or rely solely on climatic and soil parameters. Such approaches do not adequately address the physiological characteristics, growth patterns, and perennial nature of black pepper cultivation. Moreover, existing models are rarely trained on Sri Lankan datasets and therefore fail to capture local microclimates, soil conditions, varietal differences, and region-specific pest pressures.

The absence of a crop-specific, explainable yield prediction system results in several critical challenges:

- Inaccurate early-stage yield forecasts, limiting farmers' ability to plan labor allocation, irrigation, fertilizer application, and post-harvest processing efficiently.
- Undetected low-yield zones within plantations, which remain unnoticed until late growth stages, reducing opportunities for timely corrective interventions.
- Limited adoption of digital decision-support tools, as current systems often provide abstract numerical outputs without intuitive explanations or visual feedback that farmers can easily interpret.
- Inconsistent data collection and premature harvesting, which disrupt full growth-cycle data availability and reduce the reliability of AI-based prediction models.

These challenges are further intensified by the lack of AI models that integrate multimodal data sources, such as IoT-based environmental sensing and image-based analysis of pepper plants, along with Explainable AI mechanisms that clarify how environmental and physiological factors influence predicted yield outcomes.

Therefore, the core research problem addressed in this study is the need to develop a unified, IoT-driven, machine learning–based yield prediction system for black pepper cultivation in Sri Lanka that provides accurate early-stage predictions and transparent, explainable insights. Addressing this problem will enable farmers to make informed agronomic decisions, optimize resource utilization, improve yield stability, and enhance trust in AI-assisted agricultural technologies.

2. OBJECTIVES

2.1 Main Objective

The main objective of the system is to develop an integrated intelligent decision support system for smart pepper farming in Sri Lanka, combining yield prediction, disease detection, market price forecasting, post-harvest quality grading, and blockchain-based traceability to enhance productivity, transparency, and profitability for farmers and stakeholders.

2.2 Specific Objectives

The primary objective of this research is to design and develop an IoT-driven, explainable machine learning system for early-stage black pepper yield prediction tailored to Sri Lankan cultivation conditions. The specific objectives are as follows,

1. To collect and prepare a multimodal dataset for black pepper yield prediction
 - To collect high-quality images of young pepper cones at early fruiting stages using mobile devices.
 - To collect real-time environmental and soil data (e.g., temperature, humidity, soil moisture, rainfall, and light intensity) using IoT sensors deployed near pepper plants.
 - To preprocess, annotate, and synchronize image data and sensor data to create a reliable dataset suitable for machine learning model training.
2. To develop an early-stage black pepper yield prediction model using machine learning
 - To design and train Convolutional Neural Network (CNN)–based models for image-based yield prediction from young pepper cone images.

- To integrate IoT sensor data with image-based features to improve prediction accuracy and robustness.
 - To evaluate model performance under varying environmental conditions and across different pepper plants.
3. To incorporate Explainable Artificial Intelligence (XAI) for transparent yield prediction
- To apply XAI techniques such as SHAP, LIME, or Grad-CAM to explain model predictions.
 - To identify and visualize key factors influencing yield predictions, including environmental parameters and visual plant features.
 - To present explanations in a farmer-friendly manner to improve trust and understanding of AI-driven recommendations.
4. To provide intuitive visual feedback for farmers through a mobile-based interface
- To visualize predicted yield and crop condition insights using simple graphical and visual representations.
 - To enable farmers to capture plant images and receive immediate yield predictions and explanatory feedback through a smartphone application.
 - To ensure usability and accessibility for smallholder farmers with minimal technical expertise.
5. To support informed decision-making in black pepper cultivation
- To assist farmers in planning irrigation, fertilizer application, and harvest timing based on predicted yield and XAI insights.
 - To reduce uncertainty in early-stage yield estimation and improve resource allocation efficiency.
 - To bridge the gap between advanced AI technologies and real-world pepper farming practices in Sri Lanka.

3. METHODOLOGY

3.1 System Overview

The proposed system is an integrated intelligent decision support platform for smart pepper farming, designed to optimize cultivation, disease management, pricing decisions, and post-harvest quality grading. It leverages computer vision, deep learning, IoT sensors, XAI, advanced forecasting models, and blockchain to provide farmers and stakeholders with actionable insights, transparency, and improved profitability.

- **Yield Prediction and Visualization**

The yield prediction component is designed to provide early-stage, plant-level black pepper yield estimates by integrating computer vision, deep learning, IoT sensor data, and explainable AI techniques. This module addresses the limitations of traditional yield estimation methods that rely on manual observation and farmer intuition, which are often subjective and inaccurate during early growth stages.

- **Disease Detection and Health Monitoring**

The system enables early disease detection using deep learning models applied to leaf images captured by farmers. It provides localized treatment recommendations, including organic and chemical solutions. Furthermore, NDVI-based health assessment detects plant stress before visible symptoms, helping farmers take preventive actions. Smart notifications alert farmers to regional disease outbreaks and provide daily plant health tips for better crop management.

- **Market Price Forecasting and Blockchain Traceability**

This module employs Temporal Fusion Transformer (TFT) for multi-variable time-series forecasting, predicting weekly pepper prices by considering weather, production, and market factors. It offers “Sell or Wait” recommendations with confidence scores to guide farmers’ marketing strategies. Additionally, all critical data yield, disease status, price forecasts, and quality grades are stored on a blockchain to ensure tamper-proof traceability. A QR code linked to each pepper batch provides transparent product history for consumers and exporters.

- **Post-Harvest Quality Grading**

The system integrates an IoT-based device (ESP32 with load cell and moisture sensor) to measure bulk density and moisture content of pepper samples. Combined with AI-based image processing, the module evaluates visual quality factors such as color uniformity, mold presence, and adulteration. A multi-parameter scoring system classifies pepper into Grade A, B, or C, aligned with national and international standards. Farmers receive downloadable quality reports, and high-quality batches trigger buyer notifications for better pricing opportunities.

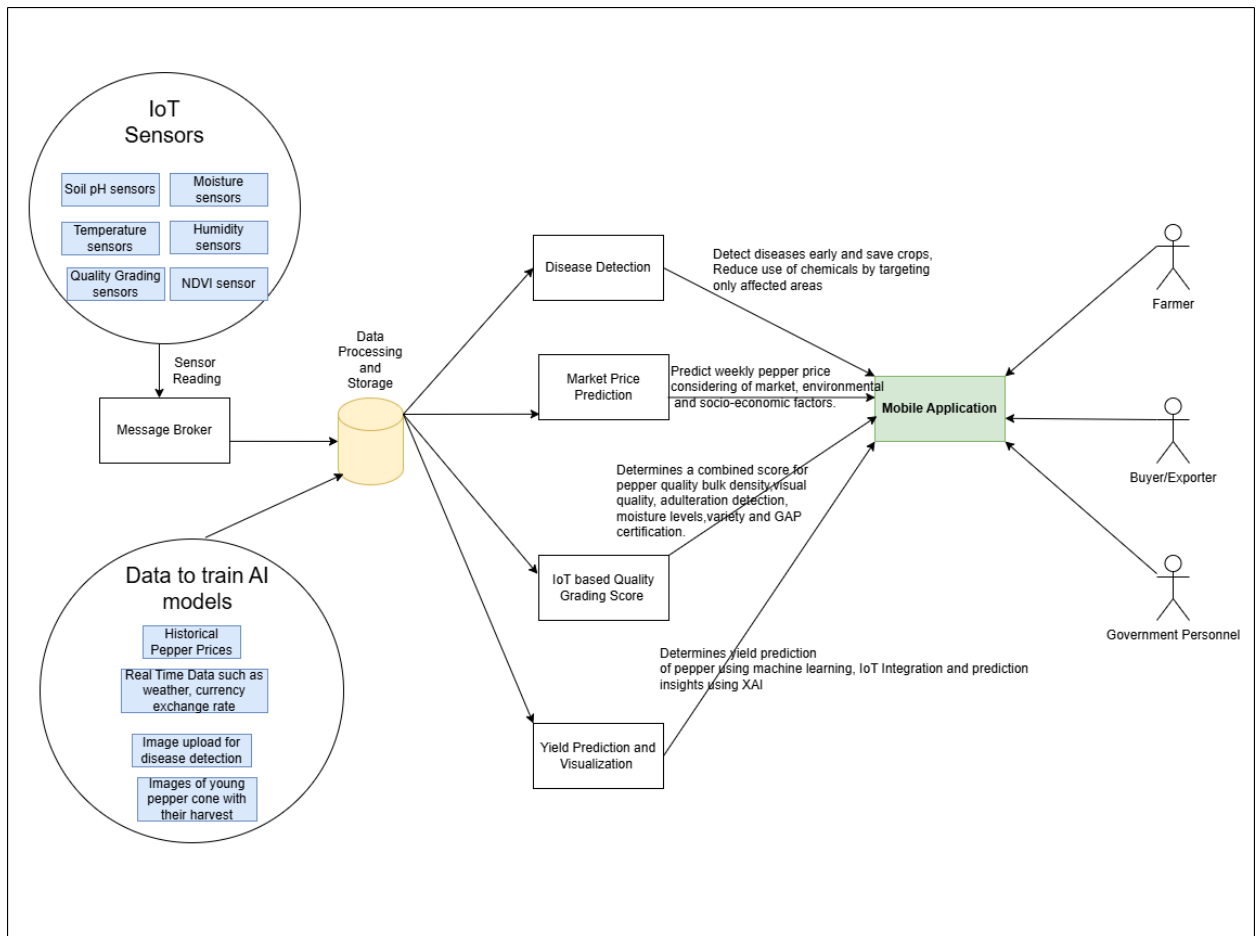


Figure 2: System Overview Diagram

3.1.1 System Architecture

The proposed system is designed as a modular, AI- and IoT-driven platform that integrates data collection, intelligent analytics, visualization, decision support, and blockchain-based traceability. It supports pepper farmers in Sri Lanka by predicting yield, detecting diseases early, forecasting market prices, grading post-harvest quality, and ensuring transparent supply chain tracking. The key Architectural Layers are,

1. Data Acquisition Layer

- IoT Devices & Sensors

ESP32-based device with load cell and moisture sensor for bulk density and moisture measurement.

- Image Capture

Farmers upload images of pepper cones (for yield prediction) and leaves (for disease detection) via the mobile app.

- External Data Sources

Weather APIs, currency exchange rates, and historical price databases for price forecasting.

2. Data Processing & AI Analytics Layer

- Yield Prediction Module

Uses CNN-based image analysis of young pepper cones to predict per-plant yield along with IoT Integration and XAI insights.

- Disease Detection Module

Deep learning (CNN and NDVI) to detect disease symptoms early.

Predicts disease progression and provides treatment recommendations.

- Market Price Forecasting Module

Uses Temporal Fusion Transformer (TFT) for multi-variable time-series forecasting.

Considers historical prices, weather, production trends, and currency rates.

- **Post-Harvest Quality Grading Module**

Combines IoT measurements (bulk density) with image-based visual grading.

Assigns Grade A/B/C based on combined score.

- **Blockchain Traceability Layer**

Records key outputs (yield, disease status, quality score, price forecasts) in a permissioned blockchain.

Generates QR codes for each batch, enabling full farm-to-market transparency.

C. Application & Visualization Layer

- **Mobile App**

Farmers can capture images, view yield predictions, and receive disease alerts.

Displays weekly price forecasts, sell/wait recommendations, and quality grading reports.

Provides community features: success stories, tips, notifications.

- **Web Dashboard**

Used by agriculture officers, buyers, and exporters to monitor production trends, quality statistics, and traceability logs.

3.1.2 Individual Component

This section focuses on the Early-Stage Pepper Yield Prediction component of the proposed system. The system is designed using a modular architecture, where each module performs a clearly defined function while remaining interoperable with other components. Such a design improves scalability, maintainability, and adaptability, which are essential characteristics of modern AI-driven agricultural decision-support systems.

• **Image Acquisition Module**

This module is responsible for collecting visual data required for early-stage yield prediction.

- To enable farmers to capture high-quality images of young pepper cones using a smartphone-based mobile application.
- To associate each captured image with relevant metadata, including timestamp, geolocation, plant identifier, and basic environmental conditions (e.g., temperature and humidity obtained from nearby IoT sensors).
- To support the creation of a labeled dataset for supervised learning and allow temporal comparison of plant development across growth stages and seasons.

This module ensures consistent and structured data collection under real farming conditions, forming the foundation for accurate yield prediction.

• **AI-Based Yield Prediction Module**

This module performs the core analytical function of the system by estimating yield at an early growth stage.

- To process early-stage pepper cone images using Convolutional Neural Networks (CNNs) or hybrid deep learning architectures optimized for agricultural imagery.
- To generate per-plant yield predictions weeks or months before harvest, enabling proactive farm planning.
- To apply transfer learning techniques to improve model robustness under varying lighting conditions, soil backgrounds, and climatic environments typical of Sri Lankan pepper plantations.
- To incorporate Explainable AI (XAI) techniques such as Grad-CAM or SHAP to highlight visual regions and features that contribute most to yield predictions, improving transparency and farmer trust.

This module bridges the gap between complex deep learning models and practical agricultural decision-making.

• **Decision Support Interface**

This module presents system outputs in a farmer-friendly and actionable manner.

- To display predicted yield values and explanatory insights through a simple, intuitive mobile dashboard.
- To provide contextual recommendations that support fertilizer planning, irrigation scheduling, labor allocation, and harvest timing.
- To ensure usability through human-centered design principles, minimizing technical complexity and promoting adoption among smallholder farmers.

The decision support interface transforms AI outputs into meaningful insights that farmers can easily understand and apply in daily operations.

3.1.3 Software Solution

The proposed software solution follows a multi-layered architecture that combines on-device intelligence with cloud–edge hybrid processing to ensure reliable operation in rural farming environments with limited or intermittent internet connectivity. The architecture is designed to support scalability, robustness, and ease of use for smallholder pepper farmers in Sri Lanka.

• **Mobile Application (Flutter Framework)**

The mobile application serves as the primary user interface between farmers and the system.

- To provide cross-platform compatibility, the application is developed using the Flutter framework, enabling deployment on both Android and iOS devices.
- To allow farmers to capture high-quality images of young pepper cones directly using smartphone cameras for yield prediction.
- To present predicted yield outputs and explanatory insights generated using Explainable AI (XAI) techniques in a simple and farmer-friendly manner.
- To synchronize collected data automatically with backend services once connectivity is restored, ensuring data consistency and continuity.

• **Backend AI Services (Python, TensorFlow)**

The backend layer is responsible for data processing, model execution, and system intelligence.

- To handle model training, inference, and periodic updates using deep learning frameworks such as TensorFlow and PyTorch.
- To process early-stage pepper cone images using CNN-based models for yield prediction.
- To deploy AI functionalities as modular microservices exposed via RESTful APIs, enabling seamless integration with the mobile application.
- To support edge-assisted or optional on-device inference, allowing basic predictions to be generated locally on the mobile device when internet access is unavailable.
- To enable future scalability, including the integration of additional modules such as disease detection, pricing forecasts, and post-harvest quality assessment.

3.1.4 Commercialization

The system holds significant potential for technology transfer and agribusiness impact in Sri Lanka and beyond:

- **Target Users**

Smallholder pepper farmers (majority of Sri Lankan farmers).

Plantation cooperatives and spice exporters.

Government extension services and NGOs.

- **Business Model**

Free tier: Basic yield prediction by weight.

Premium tier: Advanced yield prediction, historical trend analysis

B2B partnerships: Analytics and traceability solutions for exporters and certification bodies.

- **Value Proposition**

Farmers: Early insights, reduced uncertainty, better resource allocation, increased profitability.

Exporters: Reliable supply chain with predictable volumes.

Government/NGOs: Macro-level crop monitoring and policy planning.

- Scalability and Adaptability

The architecture is crop-agnostic and can be extended to other spices (cardamom, cinnamon, nutmeg, etc.).

Potential for regional/global adoption in spice-exporting economies.

3.1.5 Future Work

While the current research focuses on yield prediction , several advanced research trajectories remain open:

- Expanded Dataset & Multimodal Fusion

Incorporating multi-season, multi-location datasets to improve generalizability.

Fusion of image, sensor (soil moisture, rainfall), and remote sensing data (drones).

- Farmer Advisory & Decision Support

Personalized recommendations for irrigation, fertilizer application, and harvest timing.

AI chatbot integration for natural-language farmer queries in Sinhala/Tamil.

- Multi-Crop Generalization

Extension to other crops within the spice and horticultural domains.

Potential to form the backbone of a “Smart Spice Farming Ecosystem” in Sri Lanka.

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

Data Collection

- To allow farmers and researchers to capture and upload images of young pepper cones using a smartphone application.
- To automatically associate each image with relevant metadata, including timestamp, geographical location (GPS), plant ID, and basic environmental and climatic parameters obtained from IoT sensors or external sources.
- To store images and metadata in a structured and secure dataset suitable for supervised machine learning.
- To support optional image annotation to identify pepper cone regions for model training and validation.

Pre-Processing and Analysis

- To preprocess captured images through resizing, normalization, noise reduction, and data augmentation to improve model robustness.
- To train and deploy a Convolutional Neural Network (CNN)–based model for early-stage pepper yield prediction at the individual plant level.
- To generate yield predictions weeks or months before harvest based on visual and contextual data.
- To evaluate model performance using standard metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 score.
- To support periodic model retraining using newly collected field data to adapt to seasonal and regional variations.

Explainability and Visualization

- To provide Explainable AI (XAI) outputs that highlight influential visual and environmental features contributing to yield predictions.
- To generate visual explanations (e.g., Grad-CAM–based saliency maps) that help farmers understand how cone density, size, and distribution affect predicted yield.
- To present predictions and explanations in a simple, farmer-friendly visual format through the mobile application.

Decision Support and Reporting

- To allow farmers to view predicted yield values at the individual plant and field levels.
- To support early decision-making related to fertilizer application, irrigation planning, labor allocation, and harvest scheduling.
- To generate downloadable yield reports in user-friendly formats such as PDF or spreadsheet summaries.
- To provide notifications or alerts when predicted yields fall below expected thresholds, enabling timely interventions.

Quality Assurance and System Reliability

- To validate predicted yields against actual harvest data for continuous accuracy benchmarking.
- To ensure the system meets usability and interpretability standards through farmer-centric testing and feedback.
- To maintain data integrity, secure storage, and controlled access to sensitive farm-level information.
- To ensure reliable system performance under low-connectivity conditions through offline data caching and delayed synchronization.

4.2 Non-Functional Requirements

- **Performance:**
Generate yield predictions and visual outputs within an acceptable response time suitable for real-time or near-real-time use on smartphones.
- **Scalability:**
Support increasing numbers of users, images, and prediction requests without degradation in performance.
- **Usability:**
Provide a simple, intuitive interface designed for farmers with minimal technical knowledge.

- **Reliability:**
Operate reliably across different Android and iOS devices under varying network conditions.
- **Security and Privacy:**
Securely store Farmer data, images, and location information using authentication, authorization, and encryption mechanisms.
- **Maintainability:**
Allow easy updates to machine learning models, application features, and data pipelines.

4.3 User Requirements

- **Ease of Use:**
Farmers should be able to easily capture images, submit data, and view yield predictions with minimal interaction steps.
- **Interpretability:**
The system should present prediction results and explanations in simple, understandable formats, avoiding complex technical terminology.
- **Data Privacy:**
The system must comply with Sri Lankan data protection regulations and ensure that all farmer-generated data is securely stored and accessed only with proper authorization.
- **Support and Guidance:**
Users should have access to in-app help features, usage guidelines, and contact channels for technical support.
- **Accessibility:**
The application should function effectively in rural environments, including support for intermittent internet connectivity.

4.4 System Requirements

Hardware Requirements

Function	Hardware Requirement
For Farmers to install Mobile Application	• Smartphone • Camera resolution greater than or equal to 12MP for clear cone images. • Internet connectivity (WiFi/4G).
For Developers to AI Model Training	• Processor: Intel i7/ i5 • GPU: NVIDIA RTX 3060 (or higher) with CUDA support. • RAM: 16GB minimum. • Storage: 1TB SSD for dataset storage and model checkpoints.

Table 2: Hardware Requirements

Software Requirements

Function	Software Requirement
For development of mobile application	• Visual Studio- IDE for development • Flutter SDK- Cross platform mobile application development • Dart- Main programming language • Git, GitHub- For version controlling
AI/ML Frameworks for AI Model Development	• TensorFlow- To run CNN based yield prediction model on device • Python- For model training • OpenCV- For image processing • Jupyter Notebook- For experimentation, training and preprocessing
Backend / Data Management	• Flask / FastAPI- for model inference API. • Firebase (Authentication, Firestore, Cloud Storage) – for user accounts, image uploads, and real-time database. • SQLite – for offline image storage and prediction history.
Testing & Deployment	• Flutter Testing Framework – for unit, widget, and integration testing. • Firebase Test Lab – to test app on different devices.

Table 3: Software Requirements

5. GANTT CHART

	2025								2026				
Task	May	June	July	Aug	Sept	Oct	Nov	Dec	Jan	Feb	March	April	May
Research Topic Selection													
Topic Evaluation Form													
Feasibility Study													
Requirement Gathering													
Project Proposal													
Project Proposal Presentation													
Project Proposal Documentation													
Software Requirement Specification													
Software Designing													
Software Implementation													
Unit Testing													
System Integration Testing													
Project Status Document													
Progress Presentation 1													
Writing Research Paper													
Testing Level 2													
Progress Presentation 2													
Demo level 1													
Website Assessment													
Project Status Document 2													
Logbook													
Final Presentation and Viva													
Final Thesis													
Final Report													

Figure 3: Gantt Chart

6. WORK BREAKDOWN STRUCTURE

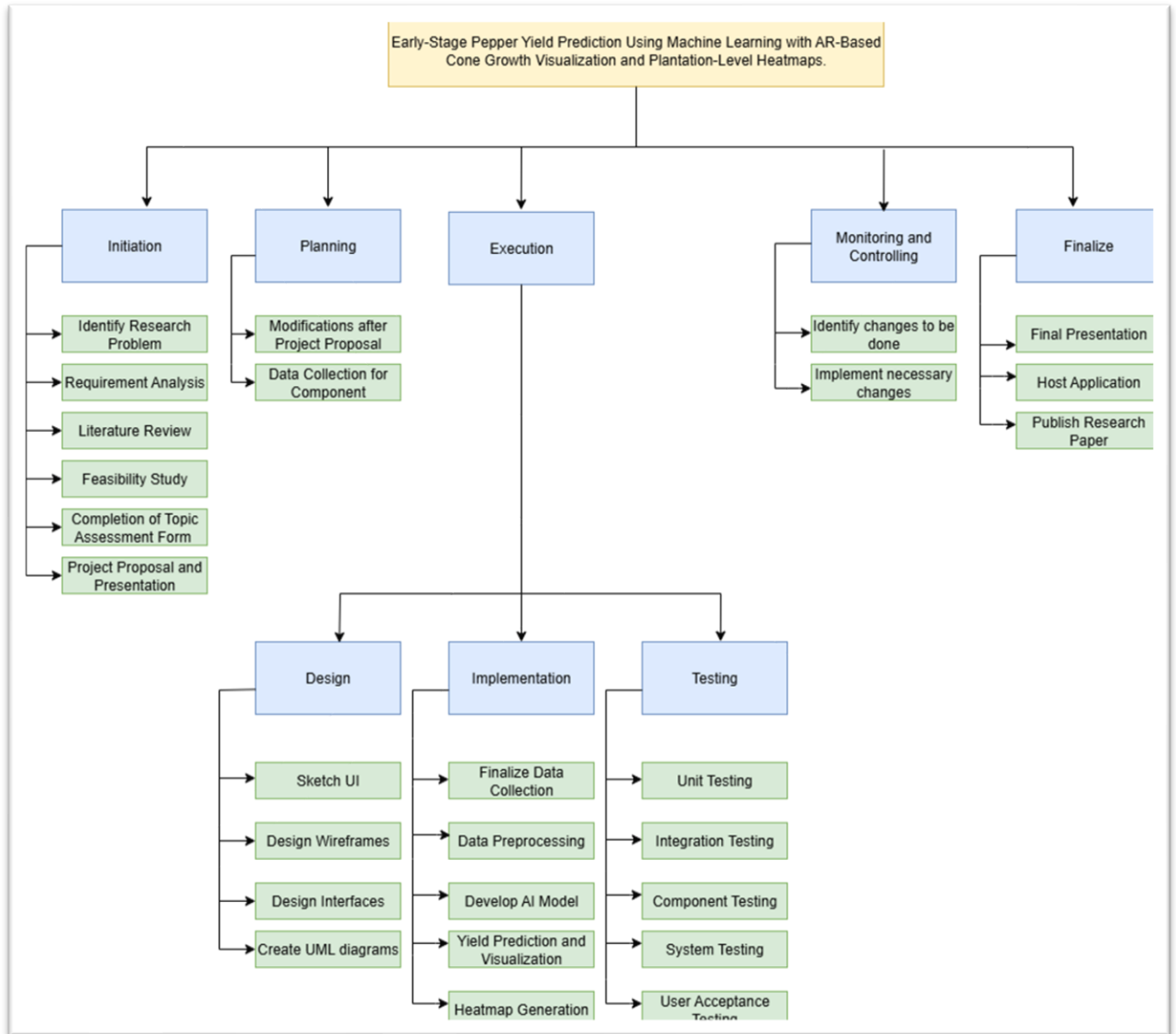


Figure 4: Work Breakdown Structure

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APPENDICES

Appendix A – Plagiarism Report

The screenshot shows a Turnitin Feedback Studio interface. The document being checked is titled "FARMING IN SRI LANKA" and is identified by ID 25-26J-479. The author is listed as "Project Proposal Report" by "Meegama D M". The document content includes the title "FARMING IN SRI LANKA", the ID "25-26J-479", and the author "Project Proposal Report Meegama D M". A specific sentence is highlighted in pink: "B.Sc. (Hons) Degree in Information Technology Specialized in Software Engineering". The plagiarism report on the right shows a total match percentage of 11%. It lists seven matches with their respective percentages and source types:

Match Number	Source	Percentage	Source Type
1	Balaji Prabhu B V, Shas...	3%	Publication
2	Submitted to Sri Lanka ...	2%	Student Paper
3	D. V. Goncharov, O. A. I...	2%	Publication
4	Submitted to National L...	1%	Student Paper
5	www.srilankabusiness...	<1%	Internet Source
6	Submitted to University...	<1%	Student Paper
7	Submitted to universitit...	<1%	Student Paper

The interface also shows a "Match Overview" tab, a "View English Sources" button, and a "Text-Only Report" option. The bottom status bar indicates "Page: 1 of 32" and "Word Count: 5364".